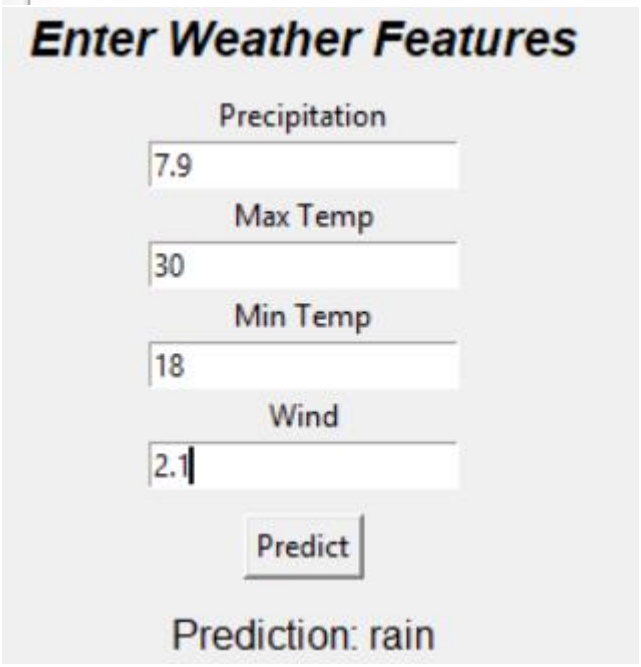
```python
# TASK 1: Weather Dataset

# train.py
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
import joblib

df = pd.read_csv("weather.csv")
df = df.drop(columns=["date"])

x = df.drop(columns=["weather"])
y = df["weather"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)

models = {
    "KNN": KNeighborsClassifier(),
    "SVM": SVC(),
    "Decision Tree": DecisionTreeClassifier()
}

best_model = None
best_accuracy = 0
best_name = ""

print("Model Accuracy")

for name, model in models.items():
    model.fit(x_train, y_train)
    y_pred = model.predict(x_test)

    accuracy = accuracy_score(y_test, y_pred)
    print(f"{name}: {accuracy*100}%")

    if accuracy > best_accuracy:
        best_accuracy = accuracy
        best_model = model
        best_name = name

print(f"Best Model: {best_name} ({best_accuracy*100}%)")

joblib.dump(best_model, "weather_model.pkl")
```

```
joblib.dump(best_model, "weather_model.pkl")
print("Model saved as weather_model.pkl")


# test.py
import tkinter as tk
import joblib

try:
    model = joblib.load("weather_model.pkl")
except:
    print("Please run train.py first")
    exit()

def predict():
    values = [
        float(prec.get()),
        float(max_temp.get()),
        float(min_temp.get()),
        float(wind.get())
    ]

    prediction = model.predict([values])[0]
    result.configure(text=f"Prediction: {prediction}")

root = tk.Tk()
root.title("Weather Prediction App")
root.geometry("350x400")

tk.Label(root, text="Enter Weather Features", font=("Arial", 14, "bold italic")).pack(pady=5)

tk.Label(root, text="Precipitation").pack()
prec = tk.Entry(root)
prec.pack()

tk.Label(root, text="Max Temp").pack()
max_temp = tk.Entry(root)
max_temp.pack()

tk.Label(root, text="Min Temp").pack()
min_temp = tk.Entry(root)
min_temp.pack()

tk.Label(root, text="Wind").pack()
wind = tk.Entry(root)
wind.pack()
```

```
tk.Label(root, text="Wind").pack()
wind = tk.Entry(root)
wind.pack()

tk.Button(root, text="Predict", command=predict).pack(pady=10)

result = tk.Label(root, text="", font=("Arial", 12))
result.pack()

root.mainloop()
```

```
========== RESTART: C:/Users/krsub/OneI
Model Accuracy
KNN: 78.15699658703072%
SVM: 76.45051194539249%
Decision Tree: 74.06143344709898%
Best Model: KNN (78.15699658703072%)
Model saved as weather_model.pkl
```

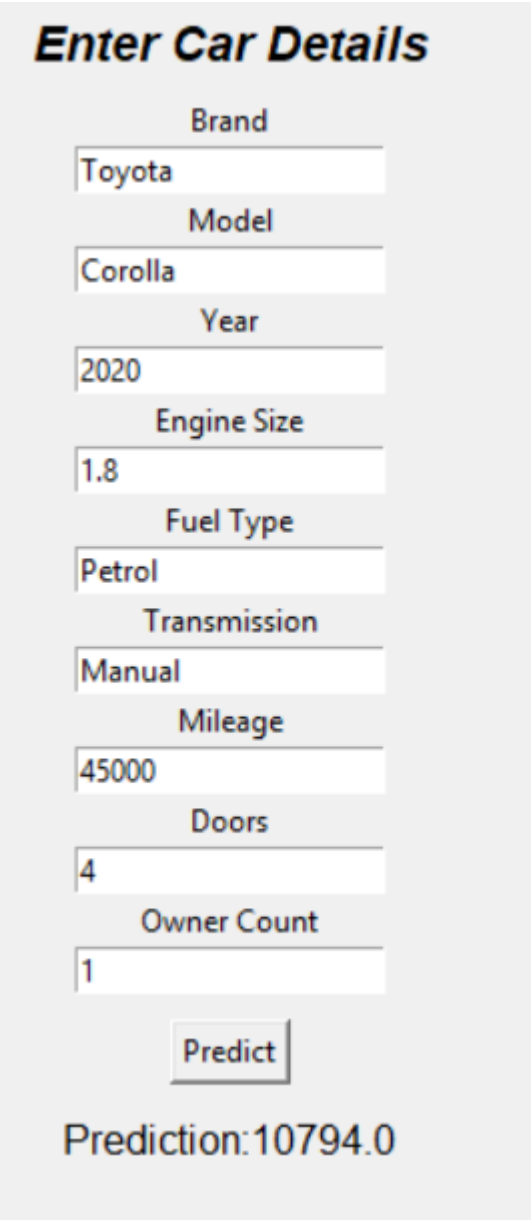


```python
# TASK 2: Car Price Dataset

# train.py
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
from sklearn.neighbors import KNeighborsRegressor
from sklearn.svm import SVR
from sklearn.tree import DecisionTreeRegressor
from sklearn.preprocessing import LabelEncoder
import joblib

df = pd.read_csv("car_price_dataset.csv")

encoders = {}
cols = ["Brand", "Model", "Fuel_Type", "Transmission"]

for col in cols:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
    encoders[col] = le

x = df.drop(columns=["Price"])
y = df["Price"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)

models = {
    "KNN": KNeighborsRegressor(),
    "SVR": SVR(),
    "Decision Tree": DecisionTreeRegressor()
}

best_model = None
best_score = 0
best_name = ""

print("Model Performance")

for name, model in models.items():
    model.fit(x_train, y_train)
    y_pred = model.predict(x_test)
```

```
    y_pred = model.predict(X_test)

    score = r2_score(y_test, y_pred)
    print(f"{name}:{score}")

    if score > best_score:
        best_score = score
        best_model = model
        best_name = name

print(f"Best Model:{best_name}({best_score})")

joblib.dump(best_model, "car_model.pkl")
joblib.dump(encoders, "encoders.pkl")

print("Model saved")


# test.py
import tkinter as tk
import joblib

try:
    model = joblib.load("car_model.pkl")
    encoders = joblib.load("encoders.pkl")
except:
    print("Run train.py first")

def predict():
    brand = encoders["Brand"].transform([b.get()])[0]
    model_name = encoders["Model"].transform([ml.get()])[0]
    year = int(yr.get())
    engine = float(eng.get())
    fuel = encoders["Fuel_Type"].transform([ft.get()])[0]
    transmission = encoders["Transmission"].transform([tn.get()])[0]
    mileage = int(mg.get())
    doors = int(dr.get())
    owner = int(own.get())

    data = [[brand, model_name, year, engine, fuel, transmission, mileage, doors, owner]]

    prediction = model.predict(data)[0]
    result.configure(text=f"Prediction:{round(prediction,2)}")

root = tk.Tk()
root.title("Car Price Prediction")
root.geometry("400x500")
```

```
tk.Label(root, text="Enter Car Details", font=("Arial", 14, "bold italic")).pack(pady=5)

tk.Label(root, text="Brand").pack()
b = tk.Entry(root)
b.pack()

tk.Label(root, text="Model").pack()
ml = tk.Entry(root)
ml.pack()

tk.Label(root, text="Year").pack()
yr = tk.Entry(root)
yr.pack()

tk.Label(root, text="Engine Size").pack()
eng = tk.Entry(root)
eng.pack()

tk.Label(root, text="Fuel Type").pack()
ft = tk.Entry(root)
ft.pack()

tk.Label(root, text="Transmission").pack()
tn = tk.Entry(root)
tn.pack()

tk.Label(root, text="Mileage").pack()
mg = tk.Entry(root)
mg.pack()

tk.Label(root, text="Doors").pack()
dr = tk.Entry(root)
dr.pack()

tk.Label(root, text="Owner Count").pack()
own = tk.Entry(root)
own.pack()

tk.Button(root, text="Predict", command=predict).pack(pady=10)

result = tk.Label(root, text="", font=("Arial", 12))
result.pack()

root.mainloop()
```

```
Model Performance
KNN:0.1575267403370686
SVR:0.17236755901644985
Decision Tree:0.9308838144537606
Best Model:Decision Tree(0.9308838144537606)
Model saved
```

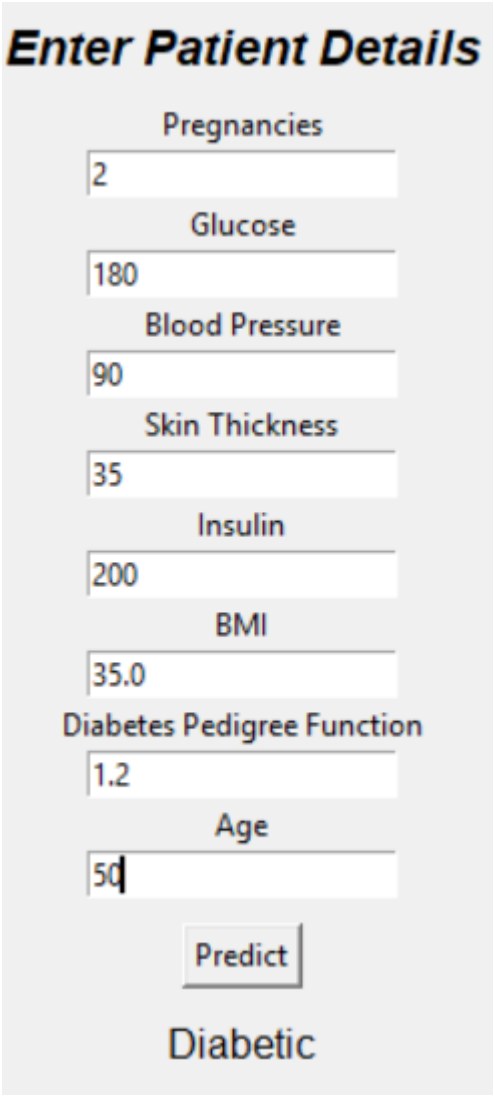

Enter Car Details
Brand
Toyota
Model
Corolla
Year
2020
Engine Size
1.8
Fuel Type
Petrol
Transmission
Manual
Mileage
45000
Doors
4
Owner Count
1
Predict
Prediction:10794.0

```python
# TASK 3: Diabetes Dataset

# train.py
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
import joblib

df = pd.read_csv("diabetes.csv")

x = df.drop(columns=["Outcome"])
y = df["Outcome"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)

models = {
    "KNN": KNeighborsClassifier(),
    "SVM": SVC(),
    "Decision Tree": DecisionTreeClassifier()
}

best_model = None
best_accuracy = 0
best_name = ""

print("Model Accuracy")

for name, model in models.items():
    model.fit(x_train, y_train)
    y_pred = model.predict(x_test)

    accuracy = accuracy_score(y_test, y_pred)
    print(f"{name}:{accuracy*100}%")

    if accuracy > best_accuracy:
        best_accuracy = accuracy
        best_model = model
        best_name = name
```

```
print(f"Best Model:{best_name}({best_accuracy*100}%)")

joblib.dump(best_model, "diabetes_model.pkl")
print("Model saved")


# test.py
import tkinter as tk
import joblib

try:
    model = joblib.load("diabetes_model.pkl")
except:
    print("Run train.py first")

def predict():
    values = [
        int(prg.get()),
        int(glc.get()),
        int(bp.get()),
        int(st.get()),
        int(ins.get()),
        float(bmi.get()),
        float(dpf.get()),
        int(age.get())
    ]

    prediction = model.predict([values])[0]

    if prediction == 1:
        result.configure(text="Diabetic")
    else:
        result.configure(text="Not Diabetic")

root = tk.Tk()
root.title("Diabetes Prediction")
root.geometry("400x500")

tk.Label(root, text="Enter Patient Details", font=("Arial", 14, "bold italic")).pack(pady=5)

tk.Label(root, text="Pregnancies").pack()
prg = tk.Entry(root)
prg.pack()
```

```
tk.Label(root, text="Glucose").pack()
glc = tk.Entry(root)
glc.pack()

tk.Label(root, text="Blood Pressure").pack()
bp = tk.Entry(root)
bp.pack()

tk.Label(root, text="Skin Thickness").pack()
st = tk.Entry(root)
st.pack()

tk.Label(root, text="Insulin").pack()
ins = tk.Entry(root)
ins.pack()

tk.Label(root, text="BMI").pack()
bmi = tk.Entry(root)
bmi.pack()

tk.Label(root, text="Diabetes Pedigree Function").pack()
dpf = tk.Entry(root)
dpf.pack()

tk.Label(root, text="Age").pack()
age = tk.Entry(root)
age.pack()

tk.Button(root, text="Predict", command=predict).pack(pady=10)

result = tk.Label(root, text="", font=("Arial", 12))
result.pack()

root.mainloop()
```

```
========== RESTART: C:/Users/krsub/OneDr
Model Accuracy
KNN:66.23376623376623%
SVM:76.62337662337663%
Decision Tree:75.97402597402598%
Best Model:SVM(76.62337662337663%)
Model saved
```


Enter Patient Details
Pregnancies
2
Glucose
180
Blood Pressure
90
Skin Thickness
35
Insulin
200
BMI
35.0
Diabetes Pedigree Function
1.2
Age
50
Predict
Diabetic